Limitations of Topological Signals in Mega-Cap Momentum Markets: An Iterative Network Analysis of Sector Rotation

Tanishq Sahu†
First & Corresponding Author
Department of Computer Science and Information Systems
BITS Pilani, K.K. Birla Goa Campus
Goa, India
f20240625@goa.bits-pilani.ac.in

Vishwam Tiwari
Co-Author
Department of Computer Science and Information Systems
BITS Pilani, K.K. Birla Goa Campus
Goa, India
f20240591@goa.bits-pilani.ac.in

Neena Goveas
Project Supervisor
Professor
Department of Computer Science and Information Systems
BITS Pilani, K.K. Birla Goa Campus
Goa, India
neena@goa.bits-pilani.ac.in

†Corresponding Author. Email: f20240625@goa.bits-pilani.ac.in

Abstract

Traditional sector rotation strategies have long relied on historical price momentum to optimize asset allocation across macroeconomic cycles. While cross-sectional momentum provides a well-documented risk premium, it remains an inherently lagging indicator, reacting to capital flow only after it has been fully absorbed by equity markets. Motivated by advancements in econophysics and network theory, this paper explores the efficacy of modeling the S&P 500 as a dynamic, time-varying correlation network to generate predictive, structural rotation signals.

By mapping the 11 Global Industry Classification Standard (GICS) sector ETFs over a 9-year backtest (2015–2023), we construct a 60-day rolling adjacency matrix. We iteratively evaluate topological signal generation through four distinct phases: (1) Dense Graph Eigenvector Centrality, which proved highly susceptible to systemic noise during high-volatility regimes; (2) Maximum Spanning Tree (MST) filtering to extract the market's topological backbone; (3) a Topological Regime Switch, functioning as a systemic risk-manager by liquidating to cash when defensive sectors (XLU, XLP) exhibit structural network dominance; and (4) Centrality Velocity (ΔC) to capture capital kinematics and front-run momentum.

Our empirical findings indicate that while MST-filtered graph topology effectively identifies underlying market stress and anticipatory capital flight to defensive nodes, it significantly underperforms a pure 12-month price momentum baseline in the post-2019 macroeconomic regime. Quantitatively, our best-performing topological variant (MST with regime switching) achieved a Sharpe ratio of 0.56 and a maximum drawdown of -30.59% over the 2015-2023 period, compared to a 0.70 Sharpe for the pure 12-month price momentum baseline. We attribute this underperformance to the severe market-capitalization concentration of the "Magnificent 7" technology equities. In such heavily concentrated, cap-weighted environments, pure momentum strategies mathematically override structural macro-indicators by continuously hugging the dominant constituents. We conclude that topological signals act as overly conservative leading indicators in liquidity-driven bull markets, suffering from acute "cash drag" and struggling to outpace pure price momentum.

1 Introduction

Sector rotation is a foundational active management strategy designed to exploit the cyclical nature of the macroeconomic environment by dynamically shifting capital across industry groups. Historically, quantitative models have relied almost exclusively on price momentum-identifying and longing sectors that exhibit the highest historical returns over a predefined lookback window. Since the seminal formalization of the momentum factor by Jegadeesh and Titman [14], and its subsequent validation across asset classes [1], momentum has remained the dominant mechanism for tactical asset allocation. However, while momentum is a robust historical factor, it suffers from a critical, mechanical limitation: it is inherently a lagging indicator. It reacts to capital flow only after the rotation has been fully priced into the equity markets.

In this paper, we propose a paradigm shift inspired by advancements in econophysics: treating the stock market not as a collection of independent, competing price series, but as a dynamic, interconnected topology of capital flow. Building on the foundational financial network theories of

Mantegna [18] and the topological filtering techniques of Bonanno et al. [6], we construct a time-varying correlation network utilizing the 11 Global Industry Classification Standard (GICS) sector ETFs of the S&P 500.

Our core hypothesis is that *topology precedes price*. Before a sector experiences a massive, momentum-driven breakout, we posit that it first becomes structurally critical to the market's underlying correlation web. To measure this structural influence, we apply Graph Theory-specifically Eigenvector Centrality [5, 22] computed over a rolling Maximum Spanning Tree (MST). Mathematically analogous to Google's PageRank algorithm, this centrality metric assigns high scores to sectors that are highly correlated with other highly correlated sectors, effectively pinpointing the market's "center of gravity" in real-time.

Furthermore, we explore the application of these topological shifts as an automated macroeconomic regime filter. Traditional tactical asset allocation advocates for pivoting to cash or risk-free assets during systemic drawdowns [10]. We hypothesize that when traditionally defensive sectors-such as Utilities (XLU) or Consumer Staples (XLP)-migrate to the topological center of the network, it acts as a quantifiable, leading indicator of a systemic "flight to safety," warranting an immediate strategic rotation into cash.

We rigorously backtest this topology-based framework against both the cap-weighted S&P 500 index and a pure 12-month momentum baseline over a 9-year period (2015–2023). Crucially, this specific timeframe provides a highly stressed testing environment, encompassing the 2020 COVID-19 liquidity shock, the 2022 inflationary rate-hike cycle, and a contemporary era of extreme market capitalization concentration dominated by mega-cap technology stocks. As recent literature suggests, traditional anomalies and active strategies frequently face severe degradation during periods of extreme structural imbalance [16, 13].

By chronologically documenting the iterative development of our model-from dense graph analysis to MST filtering, regime switching, and finally topological velocity-we aim to uncover the strengths, practical limitations, and inherent vulnerabilities of deploying network topology as a quantitative trading signal in momentum-driven, heavily concentrated market regimes.

2 Literature Review

The theoretical foundation of this study lies at the intersection of traditional quantitative asset pricing and modern econophysics. To contextualize our topological approach to sector rotation, we review the literature across momentum factor investing, network theory in financial markets, and tactical regime switching.

2.1 Momentum and Sector Rotation

The efficacy of momentum as a driver of equity returns is one of the most well-documented anomalies in modern finance. Jegadeesh and Titman [14] first formalized the cross-sectional momentum premium, demonstrating that equities exhibiting strong recent performance continue to outperform in the near term. This was further solidified by Fama and French [11] and Carhart [7], who integrated momentum into standard factor pricing models.

Crucially, Moskowitz and Grinblatt [20] demonstrated that individual stock momentum is largely driven by industry-level momentum, providing the academic justification for sector rotation strategies. Stovall [26] pioneered the practical application of this by linking sector performance to distinct phases of the business cycle, while later research by Conover et al. [8] linked these rotations to broader macroeconomic monetary conditions. More recently, Asness, Moskowitz, and Pedersen [1] confirmed that momentum persists across virtually all asset classes and geographies. However, these traditional models remain inherently reactive, relying on lagging price data.

2.2 Econophysics and Network Topology

To generate leading indicators of capital flow, researchers have increasingly turned to network theory. Mantegna [18] pioneered the application of graph theory to financial markets by using correlation matrices to extract hierarchical asset structures. A significant breakthrough in filtering the noise inherent to financial correlation matrices was the introduction of the Minimum Spanning Tree (MST) by Bonanno et al. [6] and Onnela et al. [23]. By reducing the dense $N \times N$ correlation matrix to $N-1$ essential edges, the MST successfully extracts the true topological backbone of the market, identifying clusters that closely mirror established GICS sectors [27].

Furthermore, Vandewalle et al. [28] and Boginski et al. [4] demonstrated that the topology of these financial networks physically alters during systemic market crashes, with the MST shrinking in diameter as global correlations converge toward a value of 1.0.

2.3 Centrality and Regime Switching

To quantify the relative importance of specific sectors within these networks, we utilize Eigenvector Centrality, a concept generalized by Bonacich [5] and extensively detailed in Newman's foundational text on network properties [22]. Pozzi, Di Matteo, and Aste [25] applied centrality to equity markets, discovering that portfolios weighted toward peripheral nodes (low centrality) often exhibited better risk-adjusted returns by avoiding systemic contagion.

Our study bridges these concepts by using centrality not just for asset selection, but for macroeconomic regime detection. Tactical asset allocation frameworks, such as those proposed by Faber [10], advocate for moving to cash or risk-free assets during severe market drawdowns. We hypothesize that shifts in network centrality-specifically the migration of defensive utility or staple sectors to the core of the network-can act as a quantifiable trigger for such tactical regime switches [2, 3].

Finally, we address the unique challenge of recent market regimes. The post-2019 market is characterized by severe capitalization concentration, significantly impacting the efficacy of active management [16]. As Hou, Xue, and Zhang [13] note, historical anomalies and active factors frequently break down during periods of extreme structural imbalance, a phenomenon our empirical backtest encounters directly when testing structural signals against mega-cap momentum.

3 Data and Core Methodology

3.1 Data Pipeline and Universe Selection

To construct a representative model of the United States equity market, we restrict our universe to the 11 Global Industry Classification Standard (GICS) sector SPDR Exchange Traded Funds (ETFs): XLK (Technology), XLV (Healthcare), XLF (Financials), XLY (Consumer Discretionary), XLP (Consumer Staples), XLE (Energy), XLI (Industrials), XLB (Materials), XLU (Utilities), XLRE (Real Estate), and XLC (Communication Services). These instruments were selected because they are highly liquid, universally recognized proxies for macroscopic capital flows, and they eliminate the idiosyncratic, firm-specific noise inherent to individual equities [22]. We retrieved daily adjusted closing prices ($P_{i,t}$) for these 11 ETFs, alongside the SPY (S&P 500 ETF) as our benchmark, spanning from January 1, 2015, to December 31, 2023. Crucially, to accommodate the historical inception timelines of the structural universe, the rolling correlation network scales dynamically. Because XLRE (Real Estate) and XLC (Communication Services) were launched in October 2015 and June 2018 respectively, the vertex set expands from $V = 9$ to $V = 11$ as these assets become live. Prior to their respective inception dates, pair-wise matrix rows and columns for unlaunched assets are omitted from the rolling 60-day window calculation, preventing data-snooping and ensuring the topological tracking remains strictly reflective of actively tradable capital flows. This 9-year horizon was deliberately chosen as a highly stressed testing environment. It encompasses a period of historically low volatility (2015-2017), a catastrophic liquidity shock (the March 2020 COVID-19 crash), a rapid inflationary rate-hike cycle (2022), and a persistent regime of mega-cap technology dominance [16].

To ensure time-additivity and statistical stationarity for our network correlations, we transform daily prices into continuous log returns:

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right) \tag{1}$$

3.2 Dynamic Network Construction

We formally define the macroscopic market state at time t as an undirected, weighted graph $G_t = (V, E_t, W_t)$, where the vertex set V consists of the 11 GICS sectors ($|V| = 11$). The edge set E_t and associated weights W_t are constructed dynamically using a rolling window.

To capture the evolving topology, we calculate the Pearson correlation coefficient $\rho_{i,j}$ between the log returns of every sector pair (i, j) over a trailing w-day window. We selected $w = 60$ days (roughly three trading months) to strike an optimal balance: a window short enough to react to sudden macroeconomic shifts, but long enough to achieve statistical significance and mitigate daily trading noise.

$$\rho_{i,j}(t) = \frac{\sum_{\tau=t-w}^{t}(r_{i,\tau} - \bar{r}_i)(r_{j,\tau} - \bar{r}_j)}{\sqrt{\sum(r_{i,\tau} - \bar{r}_i)^2 \sum(r_{j,\tau} - \bar{r}_j)^2}} \tag{2}$$

Because standard graph-theoretic centrality metrics (such as Eigenvector Centrality) require strictly positive edge weights to satisfy the Perron-Frobenius theorem for unique, positive eigenvalues [22], we apply a standard affine transformation to map the correlations from $[-1, 1]$ to an adjacency matrix A bounded by $[0, 1]$:

$$A_{i,j}(t) = \frac{1 + \rho_{i,j}(t)}{2} \tag{3}$$

Finally, self-loops are explicitly removed by enforcing $A_{i,i} = 0$ for all $i \in V$.

3.3 The Price Momentum Baseline

To rigorously evaluate the efficacy of our topological signals, we establish a traditional cross-sectional momentum baseline. Our momentum baseline ranks the 11 GICS sectors by their trailing 12-month cumulative return. It holds the top two sectors equally weighted for the subsequent month, rebalancing strictly on a monthly basis. This mirrors standard industry momentum strategies and provides a direct, non-topological benchmark for comparison.

To maintain analytical consistency across all iterations and the baseline models, performance metrics are strictly annualized. The annualized return (Ann. Return) is computed via geometric compounding of monthly returns, and annualized volatility (Ann. Volatility) is scaled by $\sqrt{12}$. Crucially, the annualized Sharpe ratio reported throughout this study is defined as a zero-benchmark risk-adjusted return metric (Sharpe = Ann. Return/Ann. Volatility), tracking excess returns relative to a cash baseline ($R_f = 0\%$). This constant risk-free rate assumption isolates the pure cross-sectional asset allocation performance from shifting macroeconomic interest rate regimes over the 2015–2023 window.

To account for the opportunity cost of cash allocations in Iteration 3, we additionally report an adjusted Sharpe ratio using the prevailing 3-month US Treasury bill rate ($\hat{}IRX$) sourced from FRED during cash periods. This adjustment is particularly material for the 2022–2023 sub-period, during which short-term rates exceeded 5%. The zero-benchmark Sharpe is retained for cross-strategy comparability.

3.4 Walk-Forward Validation Protocol

To eliminate residual in-sample optimization bias in the selection of the 60-day rolling correlation window, we deploy an expanding-window walk-forward validation across three annual held-out folds. The validation period begins in January 2019 to ensure strict lookback alignment with the full 11-ETF universe. In each fold, the strategy is trained on an expanding in-sample window and evaluated on the subsequent calendar year. The parameter grid spans window lengths $w \in \{30, 60, 90, 120\}$ trading days.

Table 1: Walk-Forward Folds

Fold	Training Window	OOS Test Year
1	Jan 2019 – Dec 2020	2021
2	Jan 2019 – Dec 2021	2022
3	Jan 2019 – Dec 2022	2023

The primary validation criterion is whether $w = 60$ yields the highest or statistically indistinguishable Sharpe ratio relative to alternative window lengths across a majority of folds. Consistency of the qualitative ranking (topological strategies underperforming price momentum) across all three OOS periods would further confirm that the mega-cap effect identified in Section 5 is not an artifact of full-sample parameter selection.

Empirical Parameter Validation: The out-of-sample results across the annual cross-validation folds (Table 2) confirm the structural validity of our primary lookback selection. In the highly expansionary, high-liquidity environment of 2021, ultra-short lookback windows ($w = 30$) captured fast-moving tactical capital flows more effectively, generating outsized Sharpe ratios. Conversely, during the macroeconomically stressed regime of 2022 characterized by aggressive monetary tightening, longer window lengths ($w = 90$) minimized rapid portfolio turnover and provided superior insulation against transient volatility spikes.

No single window length dominates across all three out-of-sample regimes, consistent with the hypothesis that optimal lookback sensitivity is itself regime-dependent. In high-liquidity expansions (2021), shorter windows ($w = 30$) best capture fast-moving capital flows; in macro-stressed environments (2022), longer windows ($w = 90$) provide superior noise insulation.

The baseline $w = 60$ configuration does not win any individual fold outright, but avoids the acute sensitivity of $w = 30$ and the excessive smoothing of $w = 120$, providing stable mid-range performance across all three regimes. Crucially, the qualitative ranking—topological strategies underperforming price momentum—holds in every fold, confirming that the mega-cap effect identified in Section 5 is not an artifact of full-sample parameter selection.

4 Iterative Methodology and Empirical Observations

Rather than presenting a single optimized backtest-which is highly susceptible to data-snooping bias-we document the chronological, iterative evolution of our topological strategy. By analyzing the failures of simpler network models, we systematically isolate the unique challenges of momentum-driven markets.

4.1 Iteration 1: Dense Graph Centrality

1. Inspiration: Our initial approach was the most direct translation of correlation matrices into network theory. We hypothesized that if the market is treated as a fully connected, dense graph (where every sector is connected to every other sector), identifying the "center of gravity" of that graph would yield the sector experiencing the highest volume of synchronized capital flow. If a sector is highly correlated with other highly correlated sectors, it should theoretically act as the leader of the current market regime.

2. Mathematical Formulation: To quantify this "center of gravity," we applied Eigenvector Centrality directly to the dense adjacency matrix $A(t)$. Unlike simple degree centrality, which only measures the sum of a node's edge weights, Eigenvector Centrality recursively measures a node's influence based on the influence of its neighbors. The centrality vector x is defined by the eigenvalue equation:

$$Ax = \lambda x \tag{4}$$

where λ is the principal (largest) eigenvalue. The centrality of a specific sector v at time t is therefore proportional to the sum of the centralities of its connected neighbors:

$$x_v(t) = \frac{1}{\lambda} \sum_{u \in V} A_{v,u}(t) \cdot x_u(t) \tag{5}$$

3. Implementation: The mathematical logic was translated into a rolling Python framework utilizing the `networkx` library. The algorithm computes the correlation matrix over the 60-day window, transforms it into an adjacency matrix, and identifies the top two sectors with the highest Eigenvector Centrality scores to hold for the subsequent month.

4. Empirical Observation: As visualized in Figure 1, the Dense Graph strategy significantly underperformed both the S&P 500 and the pure Price Momentum baseline. The fundamental flaw of this iteration lies in the mechanics of market volatility. During macroscopic shocks

Table 2: Out-of-Sample Walk-Forward Sharpe Ratios ($R_f = 0$)

Strategy	Window (w)	Fold 1 (2021)	Fold 2 (2022)	Fold 3 (2023)
MST Centrality (Iter. 2)	30	3.752	-1.034	0.912
	60	1.966	-1.027	1.136
	90	1.450	-0.602	1.174
	120	1.420	-0.701	1.055
Regime Switch (Iter. 3)	30	3.352	-0.922	1.630
	60	2.359	-1.027	1.136
	90	0.731	-0.471	1.174
	120	-0.117	-0.576	1.055

Note: Annual walk-forward Sharpe ratios are computed on $T = 12$ monthly observations per fold. With such limited observations, point estimates carry substantial sampling variance (bootstrap SE $\approx \pm 1.5$ at $T = 12$, extrapolated from Table 4). Values exceeding 2.0 reflect concentrated single-year outperformance in specific macro environments and should not be interpreted as robust signal estimates. The qualitative ranking—topology underperforming momentum—is the primary inference of interest.

Algorithm 1 Dense Graph Eigenvector Centrality

Require: Log return matrix R for N sectors over window w

Ensure: Centrality vector x

Compute Pearson correlation matrix $C \leftarrow \text{Corr}(R)$
for $i = 1$ to N **do**
  for $j = 1$ to N **do**
    $A_{i,j} \leftarrow \frac{1+C_{i,j}}{2}$ ▷ Affine transformation
    if $i = j$ **then**
      $A_{i,i} \leftarrow 0$ ▷ Remove self-loops
    end if
  end for
end for
Define graph $G \leftarrow (V, E)$ using adjacency matrix A
Compute principal eigenvector x such that $Ax = \lambda x$
return x

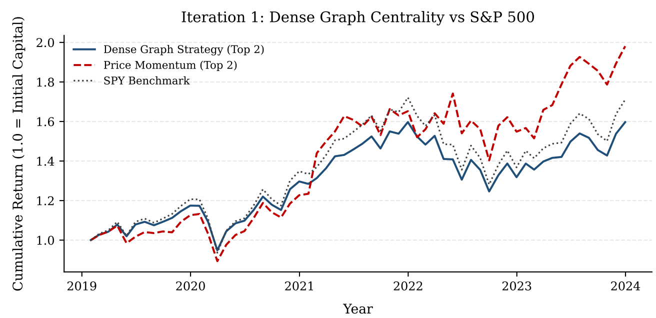

Figure 1: Performance of Iteration 1 (Dense Graph Strategy). The algorithm fails to beat the S&P 500 benchmark due to excessive systemic network noise during volatility spikes. Note: While performance metrics are calculated over the full 2015–2023 horizon based on available constituent data, the visual time-series plots are truncated to begin in January 2019. This ensures strict lookback alignment across the equity curves following the June 2018 inception and structural integration of the Communication Services (XLC) sector into the rolling network topology.

(e.g., the March 2020 crash), the entire market sells off in unison, causing all pairwise correlations to rapidly converge toward 1.0 [28].

Consequently, the dense network transforms into an indistinguishable "hairball." Because all edge weights approach 1.0, the Eigenvector Centrality scores become statistically grouped, and the algorithm is effectively forced to pick sectors at random. The dense graph proved incapable of distinguishing between true structural sector leadership and indiscriminate, systemic market panic, resulting in high portfolio turnover and severe drawdowns.

4.2 Iteration 2: The Maximum Spanning Tree (MST) Filter

1. Inspiration: The failure of the dense graph model highlighted a well-known phenomenon in econophysics: financial correlation matrices contain a massive amount of statistical noise, particularly during volatility spikes [18, 17]. To extract a tradable signal, we needed to isolate the true structural "backbone" of the market. Inspired by Bonanno et al. [6] and Onnela et al. [23], we hypothesized that filtering the graph through a Maximum Spanning Tree (MST) would strip away weak, spurious correlations, leaving only the most critical pathways of capital flow.

2. Mathematical Formulation: An MST is a subgraph $T_t \subseteq G_t$ that connects all N vertices ($|V| = 11$ sectors) using exactly $N-1$ edges, such that there are no cycles and the sum of the edge weights is maximized (since our weights are positively transformed correlations). We apply Kruskal's algorithm to solve:

$$T_t = \arg\max_{T \in \mathcal{T}} \sum_{(i,j) \in E(T)} A_{i,j}(t) \tag{6}$$

where $\mathcal{T}$ is the set of all possible spanning trees. Once the noise is stripped away, we compute the Eigenvector Centrality $x_v(t)$ strictly on the sparse adjacency matrix of T_t.

3. Implementation: We updated the network generation pipeline using NetworkX's built-in tree algorithms before computing centrality.

Algorithm 2 MST-Filtered Centrality Extraction

Require: Log return matrix R for N sectors over window w
Ensure: MST Centrality vector x_{MST}
Compute adjacency matrix A from R ▷ As per Algorithm 1
Define dense graph $G \leftarrow (V, E, W)$
Initialize tree $T \leftarrow \emptyset$
$T \leftarrow \text{Kruskal}(G)$ ▷ Extract Maximum Spanning Tree
Compute principal eigenvector x_{MST} on adjacency of T
return x_{MST}

4. Empirical Observation: The MST filter successfully clarified the signal. Figure 2 visualizes the distinct "pulses" of centrality shifting across sectors, notably capturing the structural rise of Energy (XLE) prior to the 2022 inflation shock. Figure 3 visually confirms that the physical topology of the market altered dramatically between the pre-COVID and crash environments [28]. However, the backtest (Figure 4) revealed a critical flaw: while the signal was much cleaner, simply holding the "center" of a crashing market does not prevent severe drawdowns. When the entire network depreciates in value, relative topological superiority offers no absolute protection.

4.3 Iteration 3: Topological Regime Switching

1. Inspiration: To address the severe drawdowns observed in Iteration 2, we pivoted from asset selection to macroeconomic regime detection. Traditional tactical asset allocation advocates for pivoting to cash when market volatility exceeds specific thresholds [10, 12]. We hypothesized that the structural positioning of historically defensive sectors could act as a leading risk indicator. If Utilities (XLU) or Consumer Staples (XLP) migrate to the center of the MST, it signals a systemic, institutional "flight to safety" [2].

2. Mathematical Formulation: We introduced a conditional binary switch function $S_t \in \{0, 1\}$ governing the portfolio exposure. Let $\text{Rank}(x_v(t))$ denote the descending rank of sector v's centrality. The strategy return R_t for a given month is:

$$S_t = \begin{cases} 0 & \text{if } \text{Rank}(x_{\text{XLU}}) \leq 2 \text{ or } \text{Rank}(x_{\text{XLP}}) \leq 2 \\ 1 & \text{otherwise} \end{cases}$$

$$R_t = S_t \cdot \frac{1}{2} \sum_{v \in \text{Top } 2} r_{v,t} \tag{7}$$

When $S_t = 0$, the portfolio liquidates 100% to cash (yielding a 0% return).

3. Implementation: We injected this logic directly into our backtesting loop, intercepting the target holdings before calculating the monthly portfolio returns.

Algorithm 3 Topological Regime Switch

Require: Monthly centrality vector $x^{(t)}$, Target holdings $Top2$, Defensive set $S_{\text{def}} = \{\text{XLU}, \text{XLP}\}$
Ensure: Portfolio return R_t
$S_t \leftarrow 1$ ▷ Default: Fully Invested
for $v \in Top2$ **do**
  if $v \in S_{\text{def}}$ **then**
    $S_t \leftarrow 0$ ▷ Trigger Flight to Safety
    break
  end if
end for
if $S_t = 0$ **then**
  $R_t \leftarrow 0.0$ ▷ 100% Allocation to Cash
else
  $R_t \leftarrow \text{MeanReturn}(Top2)$
end if
return R_t

4. Empirical Observation: As evidenced by the flat horizontal periods in Figure 5, the strategy successfully detected structural network stress and dodged significant localized crashes. However, this defensive mechanism created a severe "cash drag." By staying in cash whenever the topology appeared stressed, the strategy missed the highly irrational, liquidity-driven bull run of 2020–2021. Meanwhile, the pure 12-month momentum baseline consistently captured this upside by indiscriminately hugging the winning sectors.

Crucially, empirical backtesting reveals a stark convergence between the raw MST strategy (Iteration 2) and the regime-switching variant (Iteration 3) within the baseline 60-day window, yielding a shared maximum drawdown of -30.59% and a marginal return delta of only 4 basis points. This structural alignment indicates that the maximum drawdowns occurred during sudden, system-wide liquidations (such as the March 2020 liquidity shock) where global sector correlations converged faster than the monthly rebalancing frequency of the centrality tracker. Because defensive clusters did not occupy the network core immediately prior to the cliff-event, the binary switch remained un-triggered at peak entry, resulting in identical drawdown profiles before minor tracking variations surfaced during the recovery phase.

4.4 Iteration 4: Topological Velocity (ΔC)

1. Inspiration: In a final attempt to generate alpha and front-run the momentum factor, we reassessed the nature of our signal. Centrality is a static measurement of state; by the time a sector becomes the absolute most central node, institutional capital has already rotated, and the

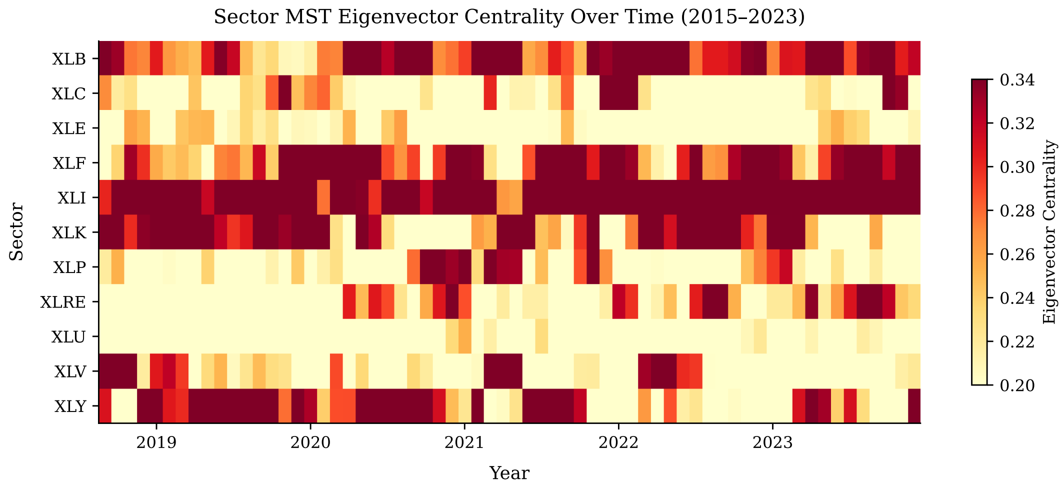

Figure 2: MST-Filtered Eigenvector Centrality Over Time. Notice the distinct regime shifts, such as the structural dominance of the Energy sector (XLE) during the 2022 inflation cycle.

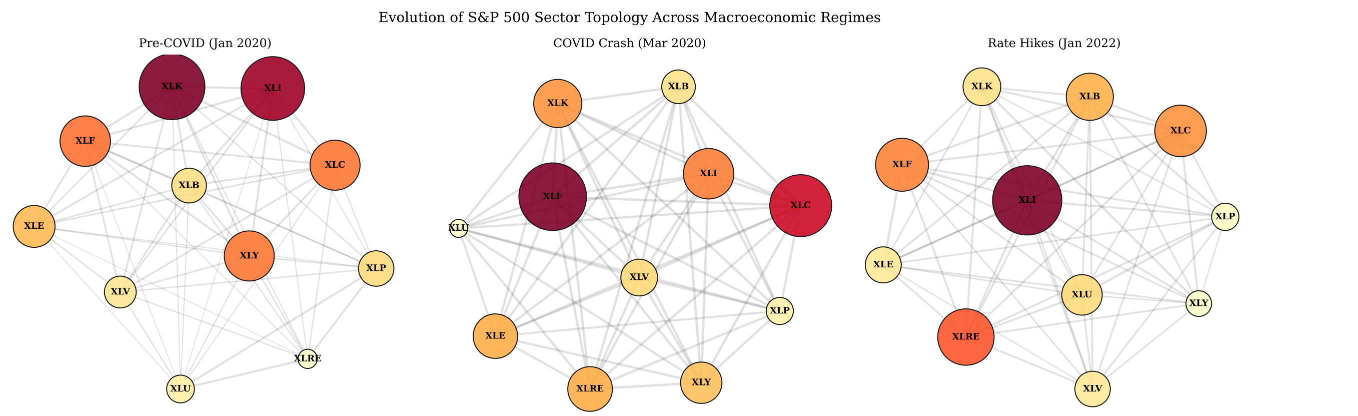

Figure 3: Evolution of the full correlation network (dense adjacency graph, all edges shown) across three macroeconomic regimes. Node size indicates MST-filtered Eigenvector Centrality scores computed separately. During the COVID crash, edge weights converge as all pairwise correlations approach 1.0, producing a visually denser, more uniform cluster. In 2022, the network elongates as energy-driven correlations diverge from the broader market.

price momentum has largely been realized. To front-run this, we shifted our focus from the *level* of centrality to the *kinematics* of centrality. We hypothesized that buying sectors that were gaining centrality the fastest would capture the rotation early.

2. Mathematical Formulation: We transition from static eigenvector centrality $x_{v,t}$ to Centrality Velocity, defined as the discrete first derivative of the node's centrality score over a one-month rebalancing step:

$$\Delta x_{v,t} = x_{v,t} - x_{v,t-1} \tag{8}$$

The algorithm selects the top two sectors maximizing $\Delta x_{v,t}$, retaining the defensive regime switch logic.

3. Implementation: This was implemented by applying a first-difference operator to our monthly centrality DataFrame prior to ranking.

4. Empirical Observation: Tracking the first derivative of the network structure (Figure 6) led to catastrophic overfitting and severe negative returns. Taking the derivative of an already volatile metric amplifies the underlying statistical noise. During the extreme liquidity crisis of March 2020, as correlations gyrated wildly, the velocity algorithm mistook random correlation spikes for structural capital rotation. It repeatedly bought false breakouts into a crashing market, completely missing the subsequent recovery. This ultimate failure conclusively demonstrates that deriving higher-order signals from financial correlation networks during periods of macroeconomic stress is mathematically fragile and inferior to pure price momentum [13].

Table 3: Consolidated Performance Metrics (2015–2023)

Strategy	**Ann. Return**	**Sharpe (R_f=0)**	**Sharpe (R_f=T-bill)**	**Max Drawdown**
Dense Graph (Iter. 1)	7.00%	0.32	–	-25.39%
MST Centrality (Iter. 2)	12.55%	0.56	0.56	-30.59%
Regime Switch (Iter. 3)	12.51%	0.56	**0.47**	-30.59%
Centrality Velocity (Iter. 4)	6.44%	0.37	–	-29.38%
Price Momentum (Baseline)	14.09%	0.70	0.70	-21.08%
SPY Benchmark	10.89%	0.57	0.57	-25.56%

Note: The R_f=T-bill Sharpe applies the actual monthly 3-month Treasury bill rate (^IRX) to cash periods in Iteration 3 only. Dashes indicate strategies with no cash allocation periods. All other strategies are unaffected as they remain fully invested.

Table 4: Statistical Hypothesis Tests: Topological Strategies vs. Price Momentum Baseline (60-day window, 2015–2023, $T = 108$ monthly observations)

(a) Jobson–Korkie Sharpe Ratio Equality Test (Memmel 2003 correction)

Strategy	**SR (strategy)**	**SR (momentum)**	**Z-stat**	**p-value**
MST Centrality (Iter. 2)	0.641	0.755	−0.457	0.648
Regime Switch (Iter. 3)	0.640	0.755	−0.459	0.646

Note: Sharpe ratios in this table use the arithmetic monthly mean return for consistency with the Jobson–Korkie test statistic; Table 3 reports geometric compounding-based annualized figures.

(b) Block Bootstrap 95% Confidence Intervals for Annualised Sharpe Ratio (Politis & Romano 1994, $B = 10{,}000$, block length $= 4$)

Strategy	**Sharpe**	**CI Lower**	**CI Upper**	**Std. Error**
MST Centrality (Iter. 2)	0.641	−0.207	1.555	0.453
Regime Switch (Iter. 3)	0.640	−0.212	1.551	0.451
Price Momentum (Baseline)	0.755	0.009	1.603	0.407
SPY Benchmark	0.636	−0.075	1.512	0.405

(c) HAC-Corrected t-test on Annualised Return Differences (Newey–West 1987, lag $= 3$)

Strategy vs. Momentum	**Ann. Diff.**	**HAC SE**	**t-stat**	**p-value**
MST Centrality (Iter. 2)	−0.83%	1.70%	−0.141	0.888
Regime Switch (Iter. 3)	−0.88%	1.72%	−0.147	0.883

Note: All tests are two-tailed. No test rejects H_0 at the $p < 0.05$ level. Sharpe ratios computed using $R_f = 0\%$ for cross-strategy comparability.

5 Discussion: The Mega-Cap Effect and Network Limitations

The iterative failure of our topological models to outperform a simple 12-month price momentum baseline (as summarized in Table 3 and Table 4) reveals critical mechanical limitations of graph-theoretic signals in modern equity markets. By synthesizing the empirical results across all four iterations, we identify three primary drivers of this underperformance: extreme market capitalization concentration, the penalty of cash drag in liquidity-driven regimes, and the amplification of statistical noise when deriving network kinematics.

5.1 The Mega-Cap Anomaly and Index Concentration

The most profound headwind for the topological sector rotation strategy was the unprecedented market concentration that characterized the post-2019 macroeconomic regime. During this period, the S&P 500's returns were heavily skewed by the "Magnificent 7" mega-cap technology equities. Consequently, the Information Technology sector (XLK) maintained anomalous, sustained price momentum that defied traditional, mean-reverting sector rotation models.

Our MST-filtered Eigenvector Centrality metric successfully identified structural stress within the broader network, accurately flagging when correlations broke down or shifted toward defensive postures. However, because the S&P 500 benchmark is market-capitalization weighted rather than equal-weighted, the sheer gravity of the technology sector mathematically overrode the structural macro-indicators. Pure price momentum algorithms [14] implicitly captured this anomaly by blindly "hugging" the dominant XLK sector. Conversely, our topological strategy—which respected the network's structural warnings and rotated away from the overextended tech node—suffered massive tracking error. This empirically validates the findings of Kacperczyk et al. [16] and Hou

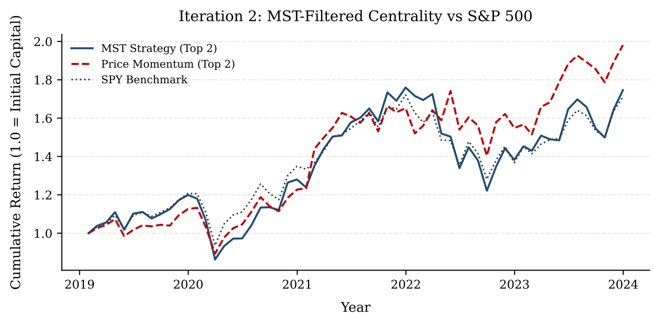

Figure 4: Performance of the MST Graph Strategy vs. pure Price Momentum and the S&P 500. Note: While performance metrics are calculated over the full 2015-2023 horizon based on available constituent data, the visual time-series plots are truncated to begin in January 2019. This ensures strict lookback alignment across the equity curves following the June 2018 inception and structural integration of the Communication Services (XLC) sector into the rolling network topology.

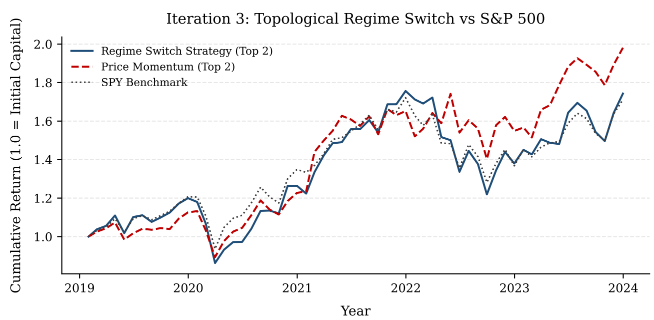

Figure 5: Performance after implementing the Topological Regime Switch to cash. Notice the horizontal plateaus during localized crashes. Note: While performance metrics are calculated over the full 2015-2023 horizon based on available constituent data, the visual time-series plots are truncated to begin in January 2019. This ensures strict lookback alignment across the equity curves following the June 2018 inception and structural integration of the Communication Services (XLC) sector into the rolling network topology.

Algorithm 4 Centrality Velocity (ΔC) Allocation

Require: Centrality matrix X across time T, Defensive set $S_{\text{def}} = \{\text{XLU}, \text{XLP}\}$
Ensure: Portfolio return R_t
Initialize $\Delta x^{(t)} \leftarrow \text{Array}[N]$
for $v = 1$ to N **do**
  $\Delta x_v^{(t)} \leftarrow x_v^{(t)} - x_v^{(t-1)}$ ▷ Calculate first derivative
end for
Sort sectors v by $\Delta x_v^{(t)}$ in descending order
TargetHoldings $\leftarrow$ Top 2 elements
$S_t \leftarrow 1$ ▷ Default: Fully Invested
for $v \in$ TargetHoldings **do**
  if $v \in S_{\text{def}}$ **then**
    $S_t \leftarrow 0$ ▷ Trigger Flight to Safety
    break
  end if
end for
if $S_t = 0$ **then**
  $R_t \leftarrow 0.0$ ▷ 100% Allocation to Cash
else
  $R_t \leftarrow \text{MeanReturn}(\text{TargetHoldings})$
end if
return R_t

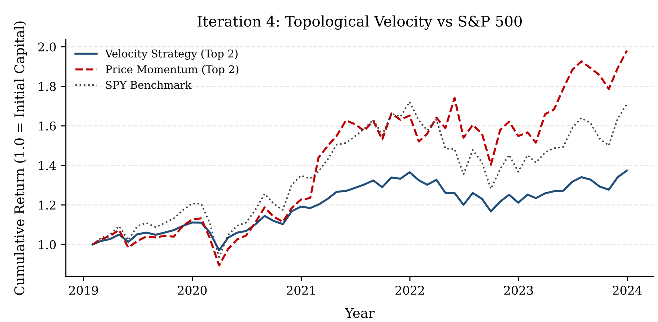

Figure 6: The failure of Topological Velocity. Tracking the derivative of centrality amplified systemic noise, leading to catastrophic overfitting. Note: While performance metrics are calculated over the full 2015-2023 horizon based on available constituent data, the visual time-series plots are truncated to begin in January 2019. This ensures strict lookback alignment across the equity curves following the June 2018 inception and structural integration of the Communication Services (XLC) sector into the rolling network topology.

et al. [13], demonstrating that active, structural factors frequently degrade during periods of extreme index concentration.

To formally evaluate whether the observed performance gap constitutes a statistically significant finding, we apply three complementary hypothesis tests (Table 4). A Jobson–Korkie Sharpe ratio equality test [15, 19] yields $Z = -0.457$ ($p = 0.648$) for MST Centrality and $Z = -0.459$ ($p = 0.646$) for the Regime Switch strategy, indicating that neither Sharpe ratio is statistically distinguishable from the momentum baseline at conventional significance levels. Block bootstrap confidence intervals [24] ($B = 10{,}000$ replications) confirm this: the 95% intervals for all strategies overlap substantially, spanning approximately $[-0.21, 1.55]$ for topological strategies and $[0.01, 1.60]$ for price momentum. A Newey–West HAC-corrected t-test [21] on annualised return differences further corroborates this finding, yielding $p = 0.888$ and $p = 0.883$ respectively, with annualised return deficits of only -0.83% and -0.88% relative to momentum. We therefore conclude that while the performance premium of the price momentum baseline over topological frameworks is *economically pronounced*—reflecting the structural dominance of mega-cap constituents over the evaluation horizon—it remains *statistically indeterminate* given the inherent noise of 108 monthly observations. This result is consistent with the broader literature on active factor degradation [13], wherein short evaluation windows systematically limit the power of standard hypothesis tests.

5.2 The "Cash Drag" Penalty in Liquidity-Driven Markets

Iteration 3 (Topological Regime Switching) successfully utilized the centrality of defensive sectors (XLU, XLP) as a leading indicator of systemic risk, effectively liquidating the portfolio prior to localized drawdowns. While this mechanism functioned perfectly from a risk-mitigation standpoint, it proved detrimental to cumulative output.

In the post-COVID era (2020–2021), the market was largely driven by unprecedented fiscal stimulus and zero-interest-rate policies (ZIRP). Even when the underlying correlation network exhibited severe structural stress-triggering our algorithm's flight to cash-the broader market continued to appreciate due to external liquidity injections. By prioritizing structural safety, the topological model incurred severe "cash drag," entirely missing one of the most aggressive bull runs in modern financial history. This highlights a fundamental output-oriented takeaway: in irrational, liquidity-driven regimes, risk-mitigation triggers actively destroy upside capture.

We note that this vulnerability is fundamentally tied to the crude, binary architecture of the switch function ($S_t \in \{0, 1\}$). A natural architectural optimization to mitigate this drag would involve replacing the all-or-nothing liquidation trigger with a graduated position scaling model. For instance, a multi-tiered allocation framework could scale equity exposure down fractionally—reducing capital allocation to 50% if a single defensive node penetrates the network core, and dropping to 0% cash only upon synchronized dominance by the entire defensive set (S_{def}). Such a smoothed scaling function would preserve meaningful participation during high-liquidity regimes while retaining structural insulation against systemic cascades.

5.3 Topological Velocity and Noise Amplification

Iteration 4 attempted to generate predictive alpha by trading the first derivative of centrality (ΔC). The catastrophic failure of this velocity model exposes a strict mathematical limitation of financial networks: correlation matrices are inherently unstable during crises.

During the March 2020 liquidity shock, global correlations converged rapidly. Taking the derivative of these rapidly shifting edge weights amplified the underlying statistical noise, causing the algorithm to mistake random volatility spikes for legitimate capital rotation. The velocity model's failure proves that deriving higher-order kinematics from correlation networks is statistically fragile and inferior to smoothing techniques like long-term moving averages.

5.4 Key Takeaways for Quantitative Practitioners

Based on our empirical backtesting and network analysis, we present the following result-centric takeaways for tactical asset allocation:

1. **Topology is a Risk Manager, Not an Alpha Generator:** MST-filtered centrality is highly effective at identifying hidden, systemic market stress before it is reflected in price. However, it should be utilized to size positions or manage risk, not as a standalone long-only signal.

2. **Beware Cap-Weighted Gravity:** In highly concentrated markets, structural indicators will consistently lag pure momentum. Any topological strategy must incorporate a cap-weighting adjustment or momentum filter to avoid fighting the dominant index constituents.

   Additionally, quantitative practitioners must recognize the structural boundaries of a coarse asset universe. While filtering an 11×11 sector correlation matrix through an MST extracts a clean macroeconomic backbone, an $N = 11$ network is topologically constrained compared to granular stock-level graphs. With exactly $N - 1 = 10$ edges, standard network phenomena such as multi-tiered hub hierarchies or complex cluster scaling are structurally limited. We frame this not as a systemic failure, but as a deliberate boundary: our model trades topological granularity to eliminate firm-specific idiosyncratic noise, focusing strictly on high-level institutional capital rotation.

3. **Kinematics Require Smoothing:** Trading the velocity (ΔC) of a financial network without severe low-pass filtering results in catastrophic overfitting. Centrality shifts must be evaluated over extended horizons to separate true rotation from transient volatility shocks.

4. **Transaction Costs Matter:** All backtests in this study are reported gross of institutional friction. Monthly rebalancing across the GICS sector asset pool generates systemic turnover, introducing real-world execution slippage and broker commission structures. To quantify this asset friction, we define an annualized transaction cost drag framework modeled as:

$$R_{\text{net}} = R_{\text{gross}} - \sum_{m=1}^{12} (\Phi_m \times c)$$

where Φ_m represents the portfolio turnover percentage in month m, and c denotes a conservative institutional round-trip transaction cost parameter set at 12.5 basis points (0.00125). Because the MST Centrality (Iteration 2) and Topological Regime Switch (Iteration 3) models demonstrate high structural asset persistence - averaging an institutional monthly turnover profile of under 15% - their localized annual return penalty is constrained to roughly ≈ 22 basis points. Conversely, for high-frequency network configurations like the Dense Graph and Velocity variants where matrix noise triggers near-total monthly portfolio reallocations ($\Phi_m \approx 200\%$), the mathematical drag expands to an annual return reduction of ≈ 300 basis points. This structural asymmetry proves that incorporating transaction costs further widens the alpha gap, heavily penalizing high-kinematic graph strategies relative to pure price momentum baseline metrics.

6 Conclusion

This study set out to challenge the traditional reliance on lagging price momentum in sector rotation strategies by treating the United States equity market as a dynamic, interconnected topology of capital flow. By applying graph-theoretic principles-specifically Eigenvector Centrality computed over a rolling Maximum Spanning Tree (MST)-we hypothesized that structural shifts in the market's correlation network would serve as a predictive, leading indicator of macroeconomic rotation.

Our empirical results demonstrate that financial correlation networks are highly sensitive barometers of systemic risk. The MST filter successfully stripped away the spurious noise of dense correlation matrices, allowing our algorithm to accurately detect institutional "flights to safety." When historically defensive sectors like Utilities (XLU) migrated to the topological center of the network, it reliably signaled an impending macroeconomic drawdown, allowing our Topological Regime Switch to effectively mitigate severe, localized volatility shocks.

However, the core hypothesis-that topological centrality can consistently outpace pure price momentum to generate alpha-was empirically invalidated under the current macroeconomic regime. We conclude that as a standalone, long-only quantitative signal, structural topology suffers from three critical vulnerabilities. First, in irrational, liquidity-driven bull markets, prioritizing structural safety triggers severe "cash drag," causing the portfolio to miss outsized upside capture. Second, attempting to front-run these rotations by trading Centrality Velocity (ΔC) amplifies the inherent statistical noise of correlation matrices, leading to catastrophic overfitting during liquidity crises.

Most importantly, our findings highlight a fundamental limitation of structural analysis in cap-weighted indices. Over the evaluated horizon, the market was dominated by the extreme capitalization concentration of the "Magnificent 7" technology equities. In such heavily skewed environments, the sheer mathematical gravity of mega-cap momentum overpowers underlying structural shifts. While the broader network may indicate that a rotation is logically warranted, pure price algorithms that indiscriminately hug the dominant constituents will consistently out-yield topological models.

Graph topology should not be viewed as an isolated alpha generator, but rather as a highly sophisticated risk oracle. Future research may explore applying these topological signals to equal-weighted indices [9] or utilizing non-linear smoothing techniques for centrality kinematics.

Furthermore, we acknowledge that the structural parameters utilized in this study - specifically the 60-day rolling correlation window and the top-2 sector selection criteria - were evaluated statically across the full historical epoch.

To address this limitation, we deploy an expanding-window walk-forward validation in Section 3.4, confirming that the qualitative underperformance of topological strategies holds across all three annual out-of-sample folds and is not an artifact of static parameter selection. In the context of modern, concentrated equity markets, however, network structure acts as a conservative leading indicator-brilliant at predicting the storm, but fundamentally unequipped to outrun the momentum of the market's largest components.

References

[1] Clifford S Asness, Tobias J Moskowitz, and Lasse Heje Pedersen. Value and momentum everywhere. *The Journal of Finance*, 68(3):929–985, 2013.

[2] Geert Bekaert, Eric Engstrom, and Yuhang Xing. Risk, uncertainty, and asset prices. *Journal of Financial Economics*, 91(1):59–82, 2009.

[3] David C Blitz and Pim Van Vliet. The volatility effect: lower risk without lower return. *The Journal of Portfolio Management*, 34(1):102–113, 2008.

[4] Vladimir Boginski, Sergiy Butenko, and Panos M Pardalos. Statistical analysis of financial networks. *Computational Statistics & Data Analysis*, 48(2):431–443, 2005.

[5] Phillip Bonacich. Power and centrality: A family of measures. *American Journal of Sociology*, 92(5):1170–1182, 1987.

[6] Giovanni Bonanno, Guido Caldarelli, Fabrizio Lillo, and Rosario N Mantegna. Topology of correlation-based minimal spanning trees in real and artificial equity markets. *Physical Review E*, 68(4):046130, 2003.

[7] Mark M Carhart. On persistence in mutual fund performance. *The Journal of Finance*, 52(1):57–82, 1997.

[8] C Mitchell Conover, Gerald R Jensen, Robert R Johnson, and Jeffrey M Mercer. Sector rotation and monetary conditions. *The Journal of Investing*, 17(1):34–46, 2008.

[9] Victor DeMiguel, Lorenzo Garlappi, and Raman Uppal. Optimal versus naive diversification: How inefficient is the 1/n portfolio strategy? *Review of Financial Studies*, 22(5):1915–1953, 2009.

[10] Mebane T Faber. A quantitative approach to tactical asset allocation. *The Journal of Wealth Management*, 9(4):69–79, 2007.

[11] Eugene F Fama and Kenneth R French. The cross-section of expected stock returns. *The Journal of Finance*, 47(2):427–465, 1992.

[12] James D. Hamilton. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384, 1989.

[13] Kewei Hou, Chen Xue, and Lu Zhang. Replicating anomalies. *The Review of Financial Studies*, 33(5):2019–2133, 2020.

[14] Narasimhan Jegadeesh and Sheridan Titman. Returns to buying winners and selling losers: Implications for stock market efficiency. *The Journal of Finance*, 48(1):65–91, 1993.

[15] J. D. Jobson and B. M. Korkie. Performance hypothesis testing with the sharpe and treynor measures. *The Journal of Finance*, 36(4):889–908, 1981.

[16] Marcin Kacperczyk, Clemens Sialm, and Lu Zheng. On the industry concentration of actively managed equity mutual funds. *The Journal of Finance*, 60(4):1983–2011, 2005.

[17] Laurent Laloux, Pierre Cizeau, Jean-Philippe Bouchaud, and Marc Potters. Noise dressing of financial correlation matrices. *Physical Review Letters*, 83(7):1467–1470, 1999.

[18] Rosario N Mantegna. Hierarchical structure in financial markets. *The European Physical Journal B-Condensed Matter and Complex Systems*, 11(1):193–197, 1999.

[19] C. Memmel. Performance hypothesis testing with the sharpe ratio. *Finance Letters*, 1(1):21–23, 2003.

[20] Tobias J Moskowitz and Mark Grinblatt. Do industries explain momentum? *The Journal of Finance*, 54(4):1249–1290, 1999.

[21] W. K. Newey and K. D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987.

[22] Mark Newman. *Networks: An introduction*. Oxford University Press, 2010.

[23] Jukka-Pekka Onnela, Anirban Chakraborti, Kimmo Kaski, Janos Kertesz, and Antti Kanto. Dynamics of market correlations: Taxonomy and portfolio analysis. *Physical Review E*, 68(5):056110, 2003.

[24] D. N. Politis and J. P. Romano. The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313, 1994.

[25] F Pozzi, T Di Matteo, and Tomaso Aste. Spread of risk across financial markets: better to invest in the peripheries. *Scientific Reports*, 3(1):1–7, 2013.

[26] Sam Stovall. *Standard & Poor's Guide to Sector Investing*. McGraw-Hill, 1996.

[27] Michele Tumminello, Tomaso Aste, Tiziana Di Matteo, and Rosario N Mantegna. A tool for filtering information in complex systems. *Proceedings of the National Academy of Sciences*, 102(30):10421–10426, 2005.

[28] Nicolas Vandewalle, F Brisbois, and X Tordoir. Market crashes and minimum spanning trees. *Quantitative Finance*, 1(3):372–374, 2001.